

Machine Learning Systems

Course Overview

Topics covered in this lecture series:

- Model training pipelines
- Distributed data parallelism
- Inference serving and latency
- Monitoring and observability

Gradient Descent

Optimization fundamentals

Batch gradient descent updates all parameters using the full dataset.

Stochastic gradient descent (SGD) uses mini-batches for faster iteration.

Learning rate scheduling affects convergence speed and stability.

Common optimizers: Adam, AdamW, SGD with momentum.

Neural Network Architecture

Building blocks for deep learning

Layers: linear, convolution, attention, normalization.

Activation functions: ReLU, GELU, SiLU.

Backpropagation computes gradients via the chain rule.

Vanishing gradients motivate residual connections and normalization.

Distributed Training

Scaling model training across devices

Data parallelism replicates the model and shards batches.
All-reduce synchronizes gradients across workers.

Pipeline parallelism splits layers across devices.
Tensor parallelism partitions individual layer computations.